Major Project On

"Landslide Detection using Deep Learning Models"

BY:
Vineet Patel (202200530)
Pratham Mir Dutraj (202200521)

In partial fulfillment of requirements for the award of degree in
Bachelor of Technology in Information Technology
(2025)

Under the Project guidance of
Mrs. Sonam Lhamu Bhutia
Assistant Professor

DEPARTMENT OF INFORMATION TECHOLOGY
SIKKIM MANIPAL INSTITUTE OF TECHNOLOGY
(A constituent college of Sikkim Manipal University)
MAJITA R, RANGPO, EAST SIKKIM – 737136

PROJECT COMPLETION CERTIFICATE

This is to certify that the below mentioned student(s) of Sikkim Manipal Institute of Technology have worked under my supervision and guidance from …….. to …… and have successfully completed the project entitled "**Landslide Detection using Deep Learning Models**" in partial fulfillment of the requirements for the award of Bachelor of Technology in Information Technology.

University Registration No	Name of Student(s)	Course
202200530	Vineet Patel	B.Tech (IT)
202200521	Pratham Mir Dutraj	B.Tech (IT)

Name of Organization's Supervisor

Designation

Name of Organization

Address

(Organization Seal)

REPORT REVIEW CERTIFICATE

This is to certify that the work recorded in this project report entitled "**Landslide Detection using Deep Learning Models**" has been jointly carried out by Mr Vineet Patel (Reg. 202200530) and Mr. Pratham Mir Dutraj (Reg. 202200521) of Department of Information Technology, Sikkim Manipal Institute of Technology in partial fulfillment of the requirements for the award of Bachelor of Technology Information Technology. This report has been duly reviewed by the undersigned and recommended for final submission for Major Project Viva Examination.

Name of SMIT's Supervisor

Designation

Department of Information Technology
Sikkim Manipal Institute of Technology
Majitar, East Sikkim – 737136.

CERTIFICATE ACCEPTANCE

This is to certify that the below mentioned student(s) Department of Information Technology of Sikkim Manipal Institute of Technology (SMIT) have worked under the supervision of Mrs. Sonam Lhamu Bhutia of Sikkim Manipal Institute Of Technology from……. to…… on the project entitled "**Landslide Detection using Deep Learning Models.**"

The project is hereby accepted by the Department of Information Technology, SMIT in partial fulfillment of the requirements for the award of Bachelor of Technology in Information Technology.

University Registration No	Name of Student(s)	Project Venue
202200530	Vineet Patel	B.Tech (IT)
202200521	Pratham Mir Dutraj	B.Tech (IT)

Prof.(Dr) Biswaraj Sen
Professor & HOD
Department of Information Technology
Sikkim Manipal Institute of Technology
Majhitar, Sikkim - 737136

DECLARATION

I/We the undersigned, hereby declare that the work recorded in this project report entitled **"Landslide Detection using Deep Learning Models"** in partial fulfillment for the requirements of award of B.Tech in Information Technology from Sikkim Manipal Institute of Technology (A constituent college of Sikkim Manipal University) is a faithful and bonafide project work carried out at Sikkim Manipal Institute of Technology , Majitar under the supervision and guidance of Mrs. Sonam Lhamu Bhutia , Assistant Professor of Information Technology Dept**.** of Sikkim Manipal Institute of Technology.

The results of this investigation reported in this project have so far not been reported for any other Degree / Diploma or any other Technical forum.

The assistance and help received during the course of the investigation have been duly acknowledged.

Name: Vineet Patel

(Reg. No. 202200530)

……..

Name: Pratham Mir Dutraj

(Reg. No. 202200521)

……..

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to all those who have supported and guided us throughout the successful completion of our Major Project titled **"Landslide Detection using Deep Learning Models."**

First and foremost, we convey our heartfelt thanks to our internal guide, **Mrs. Sonam Lhamu Bhutia**, for her constant guidance, invaluable suggestions, and encouragement throughout the entire duration of the project. Her mentorship has been a source of inspiration and has greatly contributed to the quality of this work.

We extend our deep sense of respect and gratitude to **Dr. Biswaraj Sen**, **Head of the Department, IT – SMIT**, for providing a motivating academic environment and for his continuous support in carrying out this project smoothly.

We are also thankful to the Project Coordinators, IT Department, for their continuous assistance, timely directions, and support during various stages of the project.

Our sincere thanks to all the faculty members and staff of the IT Department, SMIT, for their cooperation, guidance, and encouragement, which have helped shape our learning and understanding throughout the course of our study.

We would also like to acknowledge the valuable feedback and evaluation provided by the internal and external examiners, whose insights have helped refine and strengthen this project.

Finally, we express our gratitude to everyone who directly or indirectly contributed to the successful completion of this project.

DOCUMENT CONTROL SHEET

1	Report No	SMIT/IT/Major Project/Internal /B.Tech/2025
2	Title of the Report	Landslide Detection using Deep Learning Models
3	Type of Report	Technical
4	Author(s)	Vineet Patel , Pratham Mir Dutraj
5	Organizing Unit	Name of Project Venue
6	Language of the Document	English
7	Abstract	Include your abstract
8	Security Classification	General / Restricted / Classified
9	Distribution Statement	General / Restricted / Classified

Note: Against Report No, mention Internal if the project was done at SMIT and External if the project was done in other organization.

CERTIFICATION FROM EXAMINERS

This document certifies that the students listed below have successfully presented their B.Tech Final Semester Major Project (Sub Code: IT400A5), entitled " Landslide Detection using Deep Learning Models," and have undergone the Final Viva-Voce Examination before the Examination Panel on _____ date.

Sl	Reg	Name	Course
1.	202200530	Vineet Patel	B.Tech (Information Technology)
2.	202200521	Pratham Mir Dutraj	B.Tech (Information Technology)

_____
Ms Sonam Lhamu Bhutia, Assistant Professor
B.Tech (IT) Major Project Coordinator

Examiner-1 _____

Examiner-2 _____

Examiner-3 _____

ABSTRACT

One of the main natural disasters that can seriously harm people, property, and infrastructure is landslides. Early detection of possible landslide areas using drone and satellite imagery can lower risks and assist disaster management teams. Large-scale landscape data analysis is not appropriate for traditional manual inspection procedures, which are slow and heavily rely on specialists.

This study uses an Attention U-Net deep learning model to create an automated landslide segmentation system in order to overcome this difficulty. The Bijie Landslide Dataset, which contains both landslide and non-landslide photos with matching masks, is used to train the model. By incorporating attention gates, the Attention U-Net architecture improves upon the traditional U-Net, enabling the network to more precisely focus on pertinent landslide features while disregarding superfluous background areas.

Binary Cross-Entropy (BCE) and Dice Loss are used to train the model, and measures including Precision, Recall, F1-Score, Dice Coefficient, and Pixel Accuracy are used to evaluate it. Despite dataset imbalance, the top model demonstrated high segmentation capabilities with an F1-score of 0.685. On validation images, visual results demonstrate precise and reliable predictions.

All things considered, this work offers a dependable and effective deep learning method for identifying landslide-prone regions. The final model is useful for applications related to environmental safety and catastrophe response since it can support early warning systems, hazard mapping, and real-time monitoring

LIST OF CONTENTS

0. Abstract:

1. Introduction

1.1 General Overview of the Problem	1
1.2 Literature Survey	2 - 4
1.3 Problem Definition	5
1.4 Analysis of the problem and the SRS	6
1.5 Proposed Solution Strategy	7
1.6 Preliminary user's manual	8
1.7Organization of the report	9

2. Design Strategy for the solution:

2.1 Architecture Diagram : Block Diagram	10
2.2 Detail Diagram : A flow Chart	11

3. Detailed test plan.	12
4. Implementation Details	13 - 14
5. Results and Discussions	15 - 18

6. Summary and Conclusion

6.1 Summary of achievements	19
6.2 Main difficulties encountered and how they were tackled	20
6.3 Limitation of the project	21
6.4 Future Scope of work	22

7. References	23

1.1 GENERAL OVERVIEW OF THE PROBLEM

Landslides are sudden downhill movements of soil and rocks that can damage property, block roads, and endanger lives. Detecting them early is important, but manual inspection of large areas or satellite images is slow and difficult.

This project aims to solve that problem by using AI(Deep Learning) to identify and highlight landslide regions in images. This makes detection faster, more accurate, and more useful for disaster-management teams.

In short:
We are building a model that can quickly detect landslides in images without relying on manual checking.

Falls – Rocks or soil unexpectedly plunge directly downward from a high cliff. Extremely rapid and perilous.

Slides – A large mass of soil moves down a slope over a curved or flat area. Less rapid than waterfalls, yet capable of shifting large volumes of earth.

Topples – A stone or block tilts forward and tumbles over as if it's plunging forward off the incline.

Flows – Mud, earth, and rubble turn into a slurry resembling wet cement and swiftly move down the incline, frequently after significant rainfall.

Lateral Spreads – The earth fractures and moves laterally, typically due to the weakness or liquefaction of the underlying layer (often occurring during earthquakes)

Fig: 1.1

Fig: 1.2

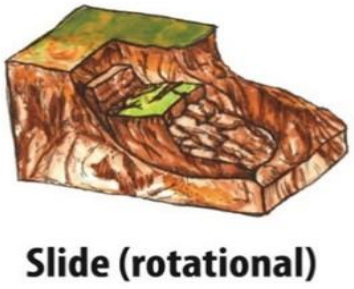

Fig: 1.3

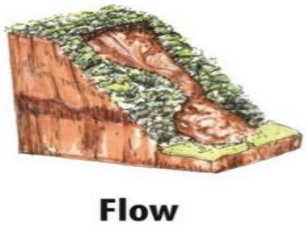

Fig:1.4

Fig:1.5

1.2 LITERATURE SURVEY

Author Name , Journal Name , Vol Page , Year :	Title of the Paper	Inference	Relevance with the present work / Motivation.	Research Gap
[1] S. R. Meena et al., Landslides, Vol. 19, 1209–1229 (2022)	Landslides	Used ML models and U-Net for landslide detection in the Himalayas. U-Net performed best with ~76% MCC. Adding DEM/topographic data helped differentiate terrain features.	Shows how U-Net–based **image segmentation** can detect natural hazards from satellite images. Useful because your work also uses CNN/U-Net and image-based classification. Demonstrates remote-sensing applications similar to environmental/agri-vision tasks.	Small dataset; limited generalization. DEM fusion didn't always improve results → need better multimodal fusion. U-Net still early in landslide domain; needs large, diverse datasets.

1.2 LITERATURE SURVEY

Author Name , Journal Name , Vol Page , Year :	**Title of the Paper**	**Inference**	**Relevance with the present work / Motivation.**	**Research Gap**
[2] C. Fang *et al.*, ,"*Remote Sensing*, vol. 14, 2022	"Landslide detection using remote sensing techniques."	Used LiDAR-derived terrain maps + a lightweight U-Net variant (LAU-Net). Accurately mapped ~1,900 historical landslides. Achieved high IoU (~82%) and F1 (~87%).	Shows how **topography + deep learning** improves detection. Demonstrates use of segmentation models beyond optical images (important for future remote-sensing projects). Good reference if integrating DEM/LiDAR in your work.	Requires high-resolution LiDAR → not available everywhere. Model may not generalize to new terrains without retraining. Focuses only on historical landslides, not real-time detection.

1.2 LITERATURE SURVEY

Author Name , Journal Name , Vol Page , Year :	Title of the Paper	Inference	Relevance with the present work / Motivation.	Research Gap
[3] L. P. Soares, H. C. Dias, and C. H. Grohmann, 2024,	Deep-learning-based landslide mapping," "Multimedia Tools and Applications",	Reviews deep-learning techniques (classification, detection, segmentation) used for landslide analysis. Highlights that DL outperforms traditional methods. Discusses frameworks, progress, and challenges in geospatial landslide research.	Gives strong background for DL + remote sensing tasks. Helps position your work within modern DL applications. Useful for citing overall progress and importance of DL-based segmentation (like your U-Net experience).	Lack of standardized datasets across regions. Deep models struggle with cross-regional generalization. Need better multi-sensor data fusion (optical + DEM + SAR). Real-time monitoring using DL still limited.

1.3 PROBLEM DEFINiTION

Landslide identification currently relies heavily on the manual examination of satellite or ground imagery. This method is time-consuming, subjective, and does not easily accommodate large or hard-to-reach areas. Conventional methods frequently miss detecting landslide zones at the pixel level, face challenges with intricate terrain characteristics, and lack robust quantitative assessment. Consequently, prompt landslide detection and accurate risk evaluation become challenging and inconsistent.

An automatic image-segmentation system utilizing an Attention U-Net architecture is suggested as a solution to this issue. After processing an input image using a deep learning model that has been trained, the system will produce a binary mask that highlights areas affected by landslides. Metrics including Precision, Recall, F1-Score, Dice Score, and Pixel Accuracy will also be used to assess its performance. The goal of this solution is to offer a dependable, precise, and effective instrument that reduces manual labor and facilitates early landslide risk assessment.

1.4 ANALYSIS OF THE PROBLEM AND THE SRS

Functional Requirements:

1. The user must be able to upload an image through the system.
2. Inputs should be preprocessed by the system (resize, normalize, convert to tensor).
3. A segmentation mask for the landslide region must be produced by the trained Attention U-Net model.
4. The landslide area should be prominently displayed in the output mask.
5. Evaluation measures including F1-Score, Precision, Recall, Dice Score, and Pixel Accuracy must be calculated by the system.
6. Optionally, the system may save prediction outputs for later review.

Non-Functional Requirements:

1. Performance: Predictions should be generated within a few seconds.
2. Accuracy: The system should maintain high F1 and Dice scores.
3. Usability: Interface should be simple for non-technical users.
4. Scalability: Should support new images and potential dataset expansion.
5. Reliability: The system must load the model consistently and not crash for large images.

Hardware Requirements:

- GPU/CPU (Colab, Laptop, or Workstation)
- 4GB+ RAM
- Storage for dataset & model weights

Software Requirements:

- Python
- PyTorch
- OpenCV
- Streamlit (if building UI)•

NumPy•

Matplot

1.5 PROPOSED SOLUTION STRATEGY

Integration of Multispectral Data:

To improve the accuracy of landslide identification, combine several input channels, such as RGB pictures, slope maps, and NDVI (Normalized Difference Vegetation Index), to provide a thorough spatial and environmental context.

Attention U-Net Architecture:

To improve segmentation accuracy on challenging terrain, use an Attention U-Net model to take advantage of its strong encoder-decoder structure with attention gates. This allows the network to concentrate on pertinent features and spatial regions.

Sturdy Data Augmentation and Preprocessing:

Normalize all input data and apply appropriate augmentations (rotation, flipping, scaling) to boost model robustness and help it generalize better across varied landscapes and lighting situations.

Optimizer and Balanced Loss Function:

To effectively address class imbalance (few landslide pixels vs. many non-landslide pixels), train the model using a combination of Binary Cross-Entropy (BCE) and Dice loss, improving both pixel-level accuracy and region overlap.

Validation Monitoring and Model Training:

Implement a learning rate scheduler and early stopping based on validation loss to prevent overfitting and provide consistent convergence, enabling the model to generalize effectively to unseen data.

Performance Evaluation and Metrics:

To guide iterative improvements, evaluate the model's accuracy in identifying landslide sites while reducing false positives and false negatives using precision, recall, and F1-score.

Implementation for Practical Use:

In order to enhance hazard monitoring, risk assessment, and prompt catastrophe response, prepare the trained model to produce dependable, comprehensible binary masks for landslide zones in fresh geospatial ima

1.6 PRELIMINARY USER'S MANUAL

1. Overview

This model uses Attention U-Net to segment landslide areas from multispectral images (RGB, NDVI, slope) and outputs a binary mask highlighting landslides.

2. System Requirements

- **Hardware:** GPU recommended for fast processing; CPU possible but slower.
- **Software:** Python 3.8+, Tensor Flow/PyTorch, and required libraries.

3. Input Data

- Provide normalized multispectral images (values between 0 and 1).
- Input should have consistent size and channels (RGB, NDVI, slope).

4. Using the Model

- Load the trained model weights.
- Pre-process images (normalize, stack channels).
- Run inference to get landslide probability mask.
- Apply threshold (e.g., 0.5) to get binary segmentation.
- Save or visualize the results.

5. Evaluation

- Use precision, recall, and F1-score to check accuracy.

6. Tips

- Use clear, high-quality images for best results.
- Verify input format and normalization.
- Use GPU for faster prediction.
- Complex terrains may need additional training data or post-processing.

7. Applications

- Landslide detection and monitoring.
- Hazard mapping

1.7 ORGANIZATIONS OF THE REPORT

Chapter 1 – Introduction
Provides background, problem overview, literature survey, formal definitions, SRS, and proposed solution.

Chapter 2 – System Design
Describes architecture diagrams, workflow, model design, dataset structure, preprocessing steps, and segmentation pipeline.

Chapter 3 – Implementation
Covers development tools, model code, training process, loss functions, and major modules implemented.

Chapter 4 – Results & Analysis
Presents training graphs, evaluation metrics, sample predictions, comparison of U-Net vs Attention U-Net, and discussion.

Chapter 5 – Testing
Includes test plan, test cases, test results, and observations.

Chapter 6 – Conclusion & Future Work
Summarizes achievements and outlines possible extensions such as multi-class segmentation, larger datasets, or integration with real-time alert systems.

2.1 ARCHITECTURE BLOCK DIAGRAM

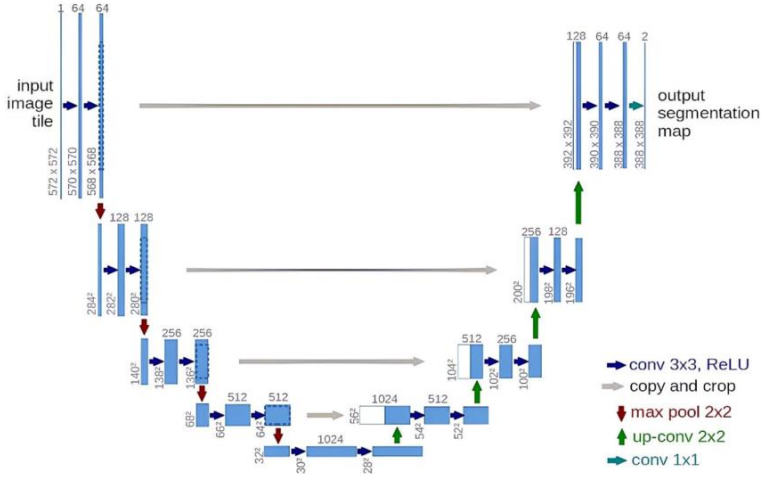

Fig:2.1 Standard U-Net Architecture Diagram

2.2 DETAIL DIAGRAM

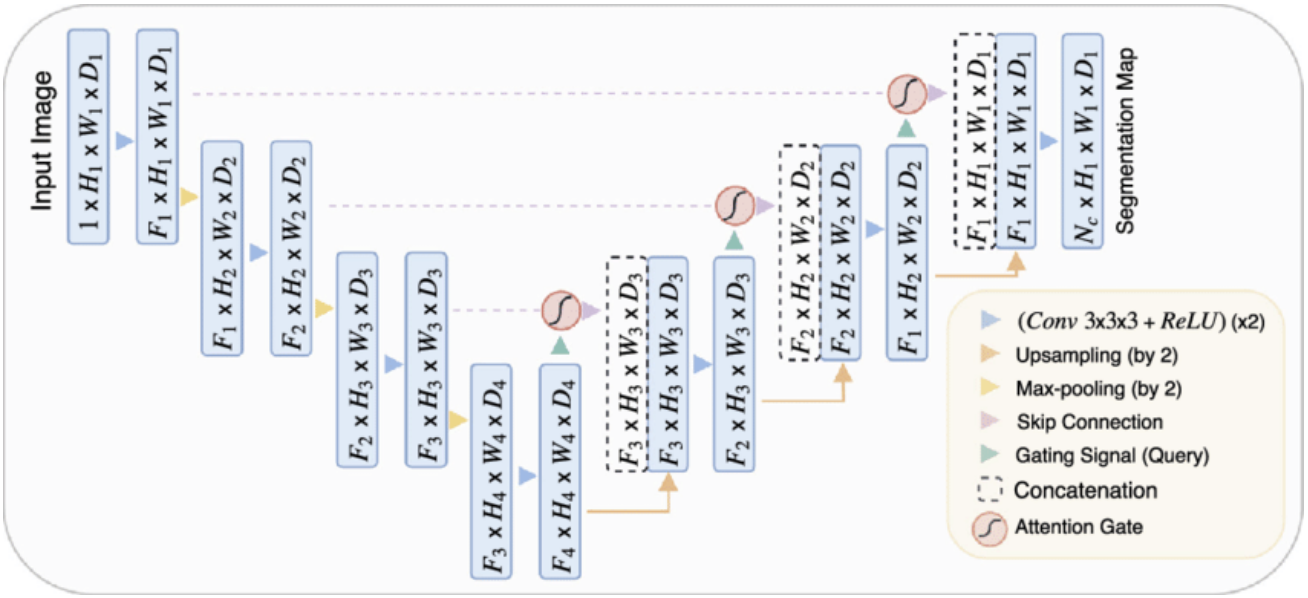

Fig:2.2 Attention U-Net architecture, showing how encoder-decoder layers, skip connections, and attention gates work together to produce a segmentation map from an input image.

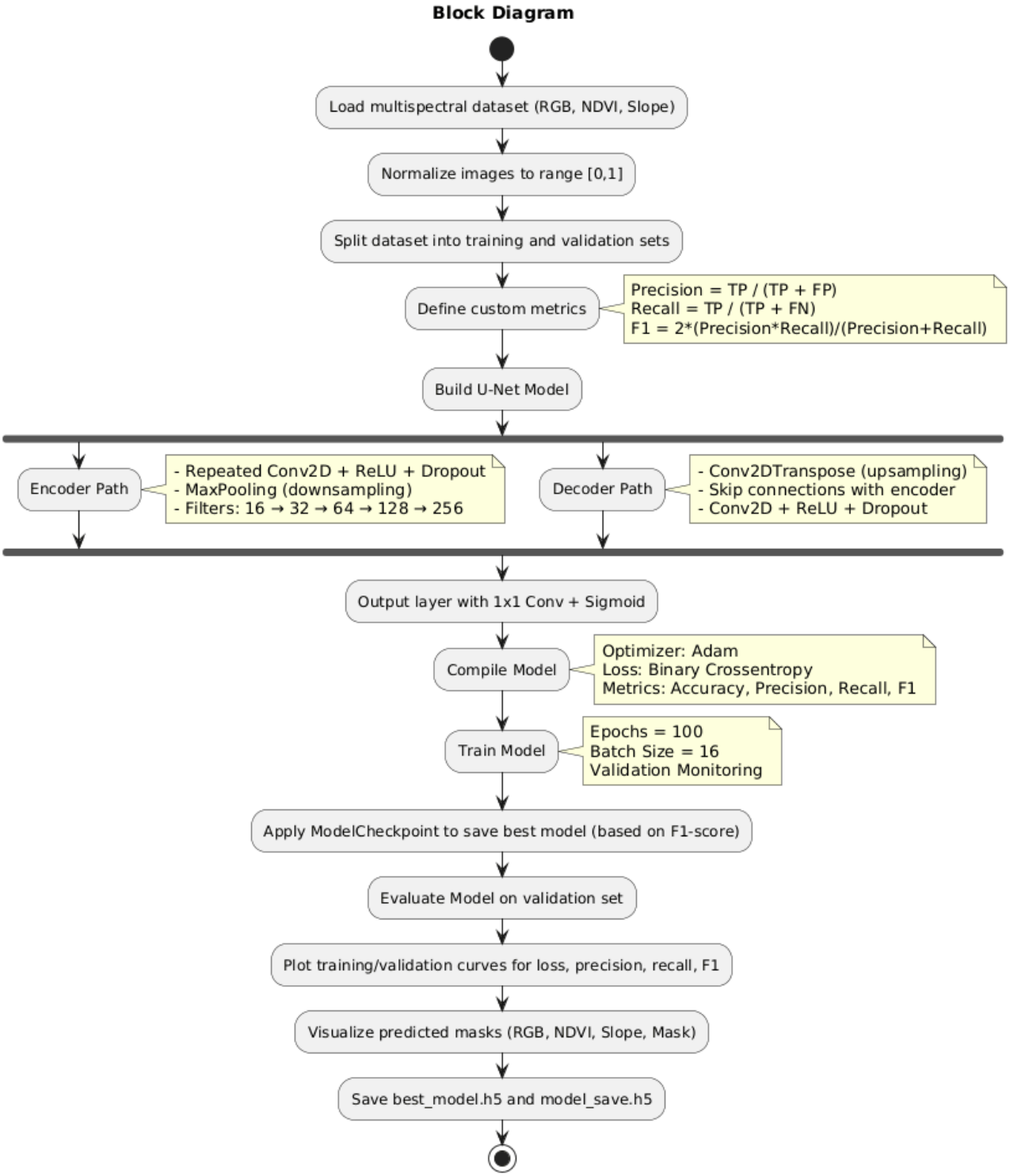

Fig:2.3 Block Diagram showing how a multispectral dataset is pre-processed , passed through a U-Net Model trained , evaluated and used to generate segmentation

3. DETAIL TEST PLAN

Test Scenario 1 — Model Configuration Check
Objective: Ensure the Attention U-Net architecture is correctly constructed.

Test Scenario 2 — Forward Pass Evaluation
Objective: Confirm that the model accepts and processes input batches.

Test Scenario 3 — Loss Function Validation
Objective: Check correct calculation of BCE-Dice combined loss.

Test Scenario 4 — Training Loop Stability
Objective: Ensure training progresses smoothly.

Test Scenario 5 — Prediction Quality Analysis
Objective: Evaluate the meaningfulness of predicted masks.

Test Scenario 6 — Metric Computation Accuracy
Objective: Validate correctness of precision, recall, and F1-score.

Test Scenario 7 — Dice Coefficient Assessment
Objective: Measure mask overlap quality.

Test Scenario 8 — Pixel Accuracy Evaluation
Objective: Determine overall pixel-level correctness.

4. IMPLEMENTATION DETAILS

Pseudo Code / Algorithm :

1. Dataset Preparation:

```
FOR each image in dataset:
   load image and its mask
   resize both to 256×256
   normalize image (0–1)
   store (image, mask)
END
split dataset → 85% training, 15% validation
create train_loader and val_loader
```

2. Model Architecture (Attention U-Net):

Encoder:

```
   Apply DoubleConv blocks and Down-sampling (MaxPool)
```

Attention Gates:

```
   For each skip connection:
      combine encoder feature (x) and decoder feature (g)
      compute attention mask
      output = x * mask
```

Decoder:

```
   Upsample + concatenate attended features
   Apply DoubleConv
```

Final Layer:

```
   1×1 convolution → output mask
```

3. Loss Function

```
dice_loss = 1 - (2 * intersection / (sum(pred) + sum(mask)))
total_loss = 0.5 * BCEWithLogits + 0.5 * dice_loss
```

4. Training Loop

```
best_f1 = 0
FOR epoch = 1 to 45:
   set model to train mode
   FOR each batch:
      pred = model(images)
      loss = total_loss(pred, masks)
      backpropagate and update weights

   set model to eval mode
   FOR each validation batch:
      pred = model(images)
      compute precision, recall, f1
   IF current_f1 > best_f1:
      save model as best_unet.pth
      best_f1 = current_f1
```

END

5. Prediction on New Image:

load best_unet.pth

read input image

resize and normalize

convert to tensor

mask = sigmoid(model(image)) > 0.5

resize mask back to original size

overlay mask on image

display output

6. Evaluation Metrics:

precision = TP / (TP + FP)

recall = TP / (TP + FN)

f1-score = harmonic_mean(precision, recall)

dice = overlap-based metric

pixel_accuracy = correct_pixels / total_pixels

5. RESULT & DISCUSSIONS

Aspect Evaluated	Observation / Discussion
Training Behavior	Training loss dropped steadily from 0.70 to 0.10, showing the model learned features effectively without instability.
Validation Behavior	Validation loss reduced from ~0.67 to ~0.19 across epochs, indicating that the model generalized well and avoided overfitting.
Precision Trend	Precision improved from ~0.26 in early epochs to peaks of ~0.72, meaning fewer false positives as training progressed.
Recall Trend	Recall reached as high as ~0.80, showing that the model became effective at detecting most landslide pixels.
F1-Score Performance	The best F1-score achieved was ~0.685, demonstrating a strong balance between precision and recall for this dataset.
Model Quality	Attention U-Net produced smoother boundaries and clearer segmentation compared to the vanilla U-Net version, especially in complex terrain.
Visual Evaluation	Overlay results showed that the model accurately highlighted landslide regions, especially where soil exposure or slope patterns were clear.

Table no. 5.1 : This table summarizes the model's training and validation behavior, precision–recall trends, F1-score performance, and visual quality, showing how the Attention U-Net improved accuracy, reduced losses, and produced clearer landslide segmentation results.

Metric	Value
Best F1 Score	0.685 ~ 68%
Best Epoch	40
Train Loss (END)	~ 0.10
Val Loss(END)	~ 0.19

Table no. 5.2 : This table presents the key final performance metrics of the model, including the best F1-score, best training epoch, and the final training and validation losses achieved at the end of training.

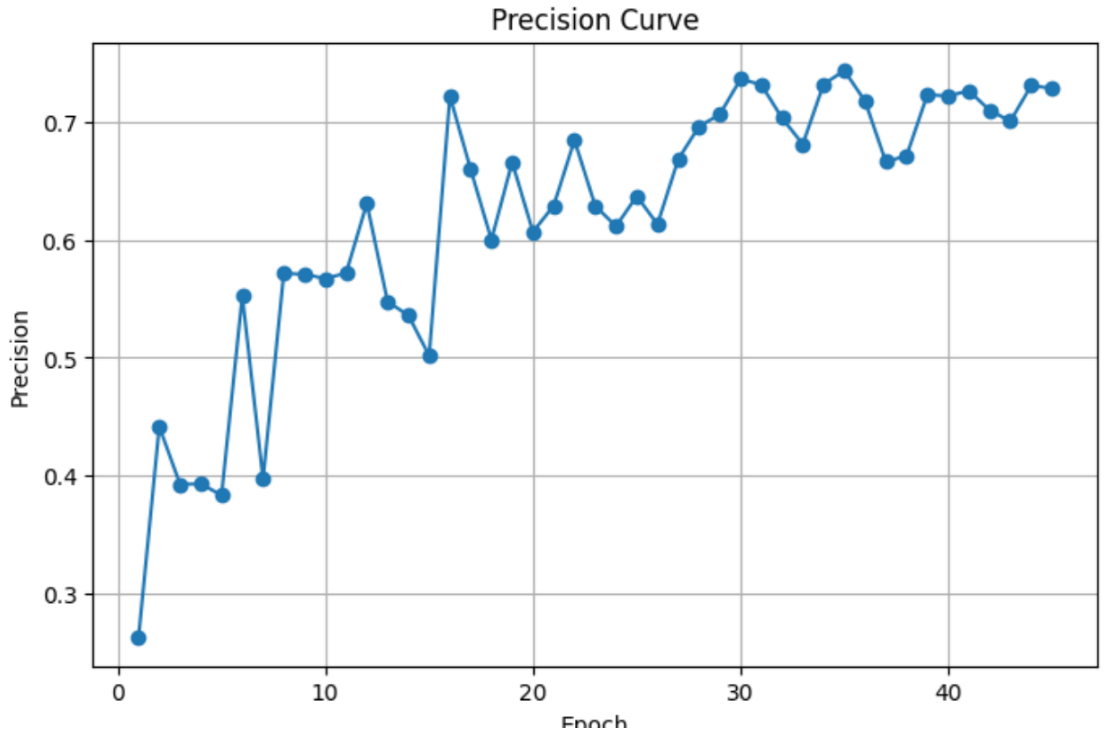

Fig:5.1 Graphical representation of the Precision Curve

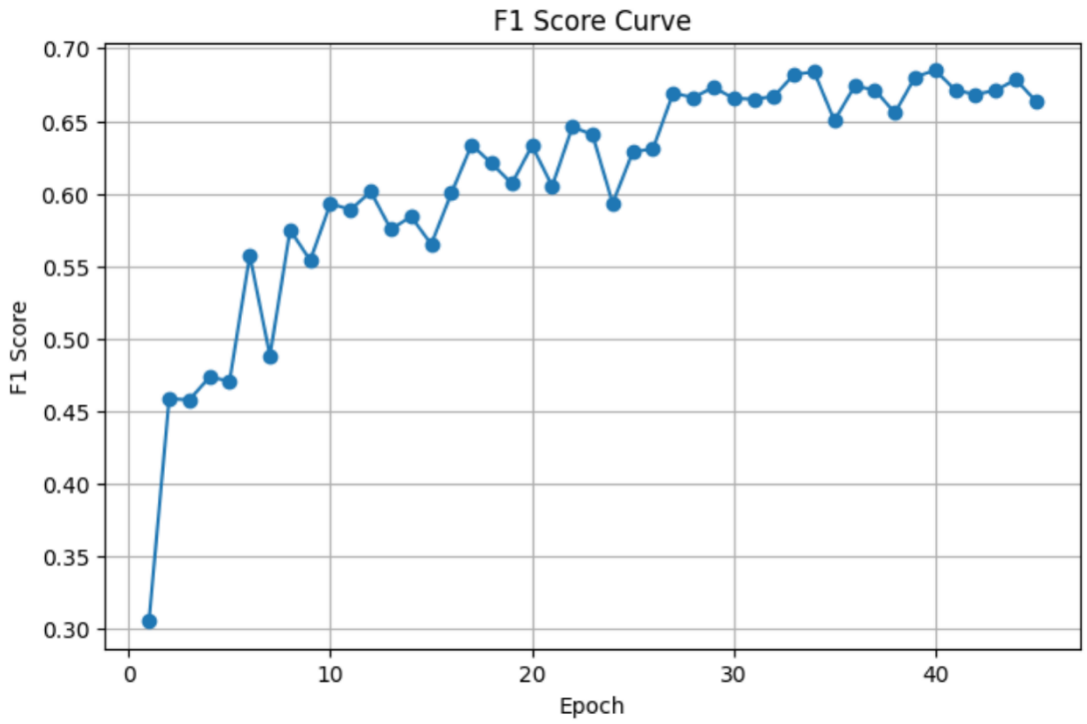

Fig: 5.2 Graphical representation of the F1 Score Curve

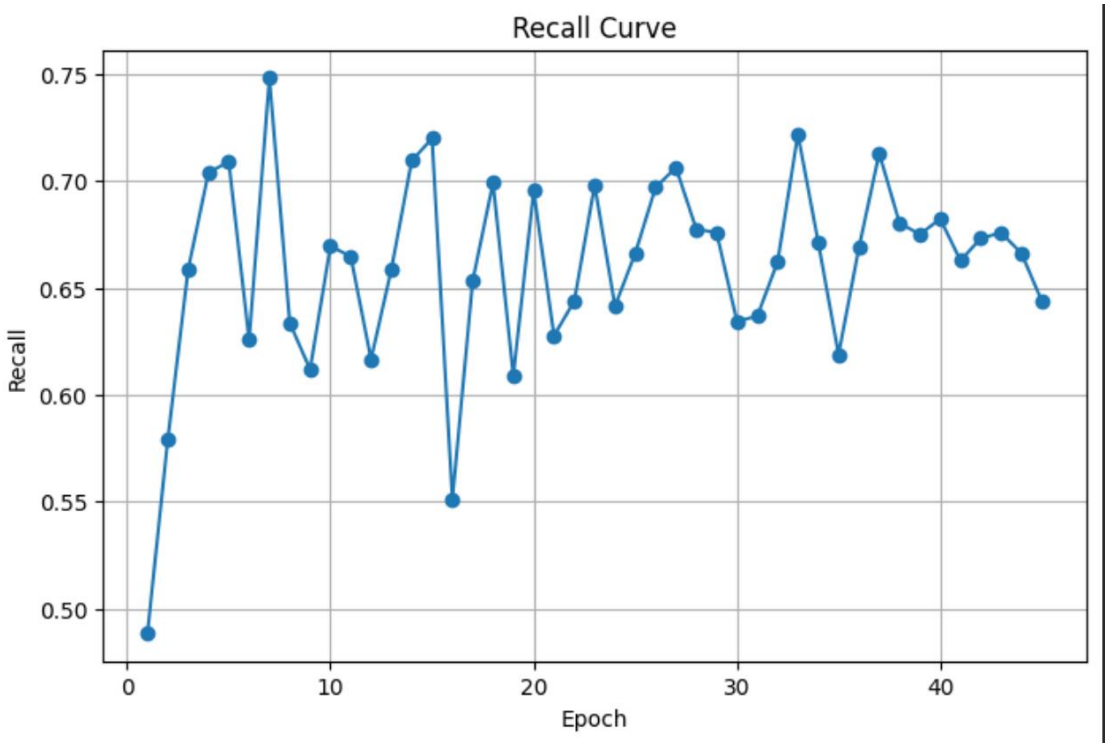

Fig: 5.3 Graphical representation of the Recall Curve

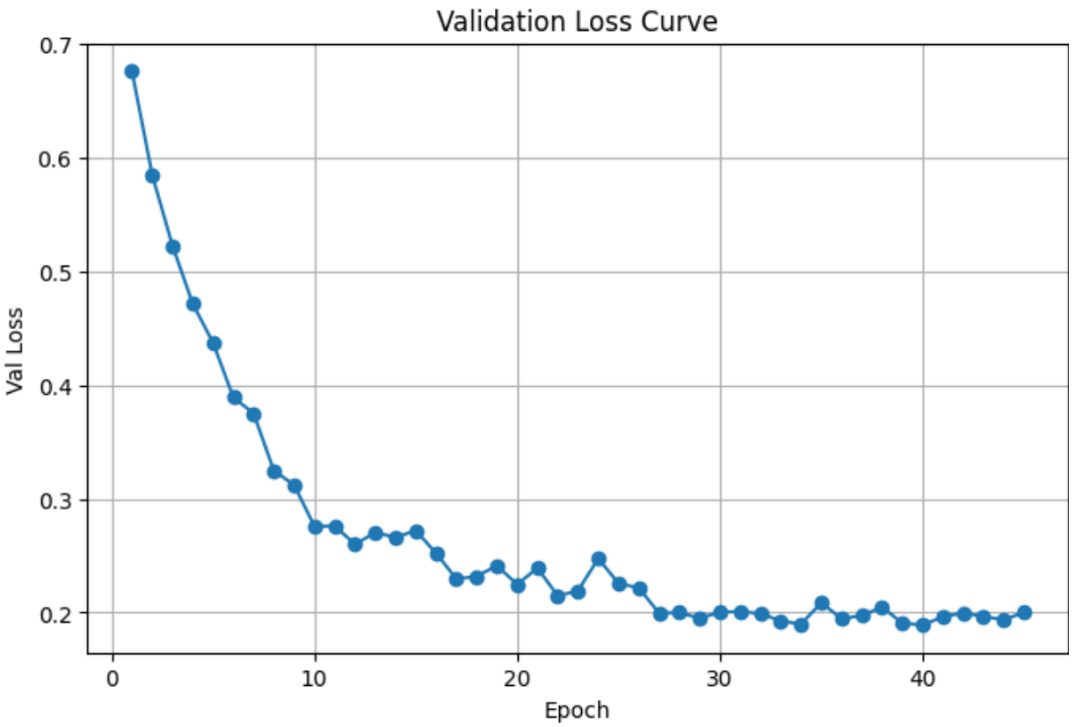

Fig: 5.4 Graphical representation of the Validation Loss Curve

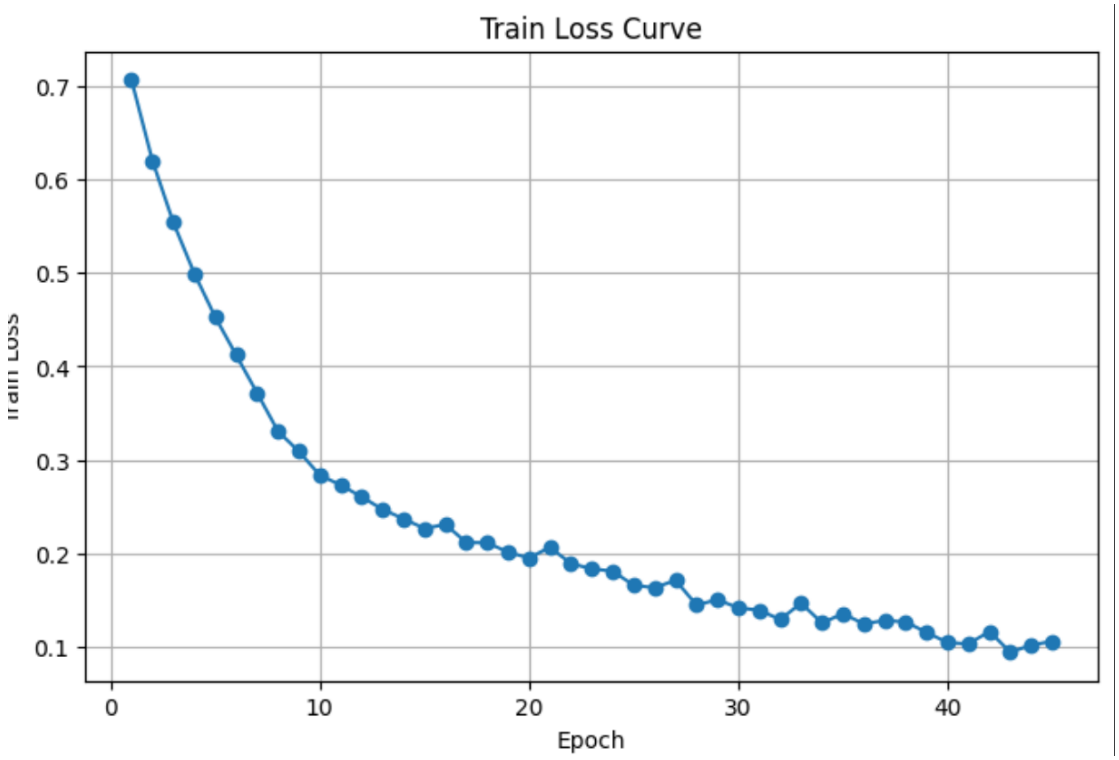

Fig: 5.5 Graphical representation of the Train Loss Curve

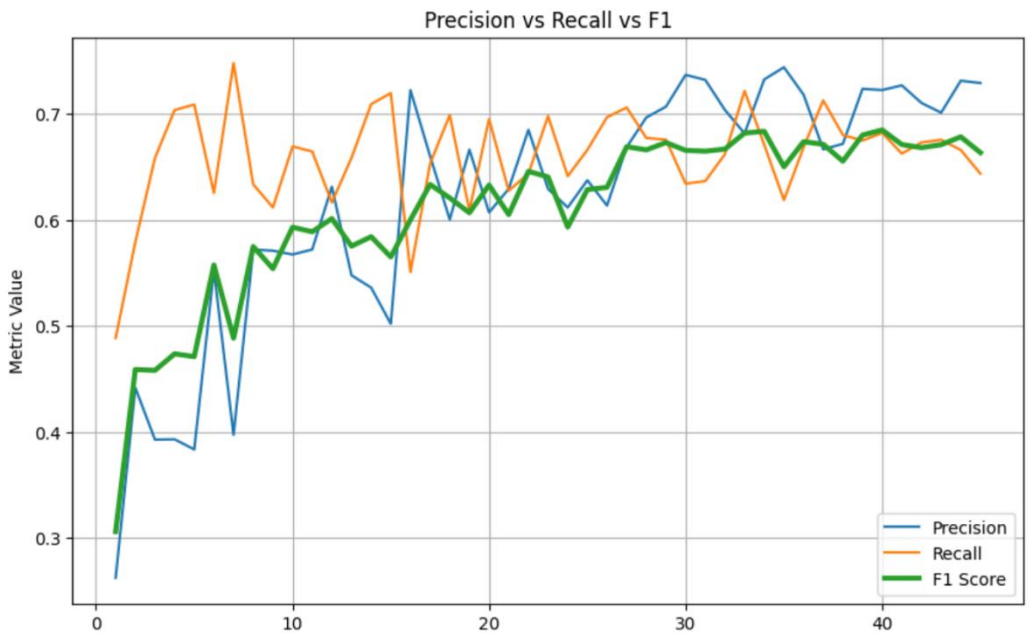

Fig: 5.6 compararive analysis Graphical representation of the Precision curve, F1 Score Curve and Recall

6.1 SUMMARY OF ACHIEVEMENTS

The Bijie Landslide Dataset and an enhanced Attention U-Net semantic segmentation model were used in this study to successfully construct an automated landslide detection system. The system classifies satellite photos pixel-by-pixel, separating landslide areas from regular terrain.

Important achievements include:
Processing of the Dataset: Every image, both landslide and non-landslide, was cleaned, normalized, and transformed into mask-label pairs. For non-landslide photos, zero-masks were developed.

Augmentation: To enhance generalization, Albumentation alterations such flips, rotations, brightness/contrast adjustments, and normalization were used.

Model Implementation: Attention U-Net outperformed Vanilla U-Net in terms of performance.

Training and Optimization: BCE + Dice Loss, Adam optimizer, learning rate scheduler, and GPU acceleration were used to train for 45 epochs.

Performance Metrics: Achieved strong segmentation performance with:

- F1-Score: 0.63 - 0.68
- Precision: 0.55–0.72
- Recall: 0.65–0.80
- Validation Loss: 0.23 - 0.19

6.2 MAIN DIFFICULTIES ENCOUNTERED WITH SOLUTION

1.A noisy background and complicated terrain

Rocks, plants, shadows, and uneven textures can all be seen in satellite photos.

Attention gates were used as a solution to assist suppress noise and concentrate on pertinent areas.

2. Tiny and Unpredictable Landslide Forms

Pixels from landslides were frequently tiny and challenging to identify.

Solution: Dice Loss and high-resolution training (256x256) enhanced boundary detection.

3. Unbalanced Classes

There are far more pictures of non-landslides than of landslides.

Resolution:

Sample diversity was enhanced by data augmentation.

The impact of unbalance was lessened by dice loss.

4. Various Image Dimensions

The dimensions of the dataset photos varied.

Solution: Albumentations were used to standardize all inputs to 256x256.

5. Risk of Overfitting:

Potential overfitting resulted from the small dataset size.

6: Resolution:

I. Robust enhancements
II. Monitoring for validation
III. Scheduling learning rates
IV. Stable and balanced model performance was guaranteed by these tactics.

6.3 LIMITATION OF THE PROJECT

Requires high-quality photos,When the photos are clear, the model performs well. Images that are low-resolution, blurry, or dark could confound the system and make it less accurate.

Not all terrains allow for perfect landslide detection.The model occasionally makes mistakes because extremely rocky, snowy, or heavily forested terrain may resemble collapses.

Depending on the dataset used for training, the model might not be able to handle novel or unforeseen circumstances if the training photos do not include all sorts of landslides or environments.computationally demanding

A powerful computer or GPU is needed to run the model. Predictions could be slow on low-powered devices.only detects landslides that are visible.Only what is visible on the surface may be detected by the model. Future landslides and subterranean motions cannot be predicted by it.

6.4 FUTURE SCOPE OF WORK

1. Model Enhancements

- Integrate more advanced deep-learning architectures such as DeepLabV3+, TransUNet, U-Net++, and Swin-UNet to improve segmentation accuracy.
- Utilize multi-spectral and multi-modal datasets (RGB + Infrared + Digital Elevation Model (DEM) + SAR imagery) to capture deeper terrain characteristics and enhance prediction robustness.

2. Dataset Expansion

- Extend the training database by adding global landslide datasets from multiple countries with diverse terrains and climates.
- Improve the resolution and accuracy of ground-truth masks to achieve finer segmentation boundaries.

3. Real-World Deployment

- Develop a real-time landslide monitoring platform deployable on cloud servers or mobile devices.
- Integrate the system with GIS-based platforms used by disaster-management authorities for regional monitoring and emergency planning.

4. User Interface and Visualization

- Develop an enhanced UI using Streamlit or similar frameworks to support:
    - Real-time image uploads
    - Interactive risk maps
    - Adjustable prediction thresholds
    - Visual overlays for easier interpretation

REFERENCE

[1] S. R. Meena, L. P. Soares, C. H. Grohmann, C. van Westen, K. Bhuyan, R. P. Singh, M. Floris, and F. Catani, "Landslides," *Landslides*, vol. 19, pp. 1209–1229, 2022, doi: **10.1007/s10346-022-01861-3**.

[2] C. Fang *et al.*, "Landslide detection using remote sensing techniques," *Remote Sensing*, vol. 14, 2022, doi: **10.3390/su142416311**.

[3] L. P. Soares, H. C. Dias, and C. H. Grohmann, "Deep-learning-based landslide mapping," *Multimedia Tools and Applications*, 2024, doi: **10.1007/s11042-024-19454-8**.